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INDEX

1 what is an index in SQL?

An index is a database object that helps retrieve data from a table faster. It improves the speed of searching from records

Think of a book. If we want to find a topic we can use the index at the back instead of reading every page

Similarly, a SQL index helps the database to find the required records without scanning the entire table in many cases

2 Example : Why is an index faster?

Imagin we have the employee table

emp_id	ename	salary
1	Arun	30000
2	Priya	40000
3	Rahul	25000
4	Divya	50000
5	Kiran	35000

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Suppose we want to find employees earning ₹ 40000

```
SELECT * FROM employee
WHERE salary = 40000;
```

without an index: The database may need to check many rows to find the matching salary

with an index on salary: The database can use the index to locate matching records more efficiently

3 Types of Indexes

i Clustered Index

Organizing table data according to the clustered index key. MySQL's InnoDB engine The primary key is normally the clustered index

Ex: emp-id is the primary key

```
CREATE TABLE employee (
emp_id INT PRIMARY KEY,
ename VARCHAR (50),
```

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Salary INT
);

ii Non clustered Index
Maintains a separate index structure that helps locate the corresponding table records
Ex: Create an index on ename

```
CREATE INDEX idx_ename
ON employee (ename);
```

4 SQL Index Syntax

a Create an Index
create an index on a specified column

Syntax: CREATE INDEX index_name
ON table_name (column_name);

Ex: CREATE INDEX idx_ename
ON employee (ename);

b View Indexes
Displays the indexes associated with a table in ~~MSL~~ MySQL

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Syntax: SHOW INDEX FROM table_name;
Ex: SHOW INDEX FROM employee;
Result: MySQL displays information about the indexes on the employee table, including their names and indexed columns

c Delete an Index
Removes an index from a table without deleting the tables data

Syntax: DROP INDEX index_name
ON table_name;
Ex: DROP INDEX idx_ename
ON employee;
Result: The index idx_ename is removed. The employee table and its row remain unchanged

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Advantage	Disadvantage
• Faster data retrival • Improves many WHERE Searches • Helps locate matching records efficiently	• uses extra storage • can show down INSERT operation • Can slow down UPDATE and DELETE operations when indexed values or entries must be maintained